AI Training Data Broker

For privacy, compliance and data teams, and owners sharing company data

The data de-identification checklist

Remove the personal information, test what is left, and keep the record your counsel will ask for.

Use it before company or health data goes to an AI company, a software vendor, a partner or a researcher. Each check names what to look for, where it hides in email, chat, tickets and records, and what to do with it.

18

Kinds of identifiers HIPAA's Safe Harbor method requires you to remove.

87%

Of Americans can be identified by ZIP code, birth date and sex alone (Latanya Sweeney, 2000).

0.09

The chance per record the European Medicines Agency accepts when it publishes clinical trial data.

Covers HIPAA Safe Harbor and Expert Determination, California's CCPA and CPRA, other US state privacy laws, GDPR and UK GDPR, and Canada's PIPEDA and Quebec Law 25.

Inside this checklist

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Jimmy Lin, AI Training Data Broker

AITrainingDataBroker.com +1 656 800 9088

1 Pick the rule you are meeting

Each privacy law sets its own test for "de-identified". Decide which ones apply before you change a single field.

The law follows the data and where the people in it live, not where your company is. A US company with European customers can fall under HIPAA, CCPA and GDPR at once. Clean to the strictest test that applies.

Rule	De-identified means	What you keep on file
HIPAA Safe Harbor 45 CFR 164.514(b)(2)	All 18 listed identifiers are removed (chapter 3 and 4), and you have no actual knowledge that what is left could identify a person.	The list of identifiers found and what was done to each.
HIPAA Expert Determination 45 CFR 164.514(b)(1)	A qualified statistical expert finds the risk of identifying anyone "very small" for a named recipient and use. Lets you keep detail Safe Harbor removes, such as exact ages or shifted dates.	The expert's signed report: method, risk found, and the conditions it holds under.
CCPA and CPRA Cal. Civ. Code 1798.140(m)	Data that cannot reasonably be linked to a consumer, plus three duties: reasonable safeguards, a public promise not to re-identify, and the same promise in every contract that passes it on.	The public statement and the contract clause.
GDPR and UK GDPR Recital 26, Article 4(5)	Anonymous: no one can identify a person by any means reasonably likely to be used. Pseudonymised data, with names swapped for codes and a key kept somewhere, is still personal data.	The test you ran and proof the key was destroyed.
Canada PIPEDA, Quebec Law 25	Quebec requires anonymisation to be irreversible by generally accepted best practices. Federal law asks whether a person can be identified.	The method, for your privacy officer.

Before you start

- [ ] **Write down who will receive the data** and what they will use it for. Expert Determination and the GDPR test both depend on it.
- [ ] **List every law that applies,** from where the people in the data live (customers, staff, patients, and the people they wrote to), and ask your counsel to confirm the list.
- [ ] **Choose Safe Harbor, Expert Determination, or both.** Safe Harbor for direct identifiers plus Expert Determination for the dates or places the use needs is allowed and common.

Write down every system, whose information is in it, and where it hides, before anything is exported.

- [ ] **List each system** the data comes from: email, chat, help desk, CRM, accounting, file shares, health records, call recordings.
- [ ] **Note the years each holds** and roughly how many records or messages.
- [ ] **Name whose information is in it:** customers, patients, staff, job applicants, suppliers, and people copied on email.
- [ ] **Mark the free text:** message bodies, ticket notes, comments and clinical notes. Personal details sit there with no field to label them.
- [ ] **Decide what stays out entirely:** staff and HR files, children's data, card numbers, call audio, anything a contract forbids sharing.
- [ ] **Export a copy to a locked workspace** that only the people cleaning it can reach. Log every access, and write down the date raw copies will be deleted.

Worksheet: data map

System	Years	Whose data	Where they live	Free text?	In, out, or in with limits
Example: help desk	2017 to 2025	Customers, staff	US, Canada	Yes, notes	In; attachments out

3 Direct identifiers

Anything that names or reaches a person comes out, or becomes a placeholder, under every rule.

Identifier	Where it hides in company data	Action	Done
Names of people, relatives, employers	Greetings, signatures, From and To lines, @mentions, ticket assignees, file names	Consistent placeholder	☐
Street address, city, county	Signatures, shipping fields, job sites, letterheads	Remove; see chapter 4 for ZIP	☐
Phone and fax numbers	Signatures, call logs, CRM fields, typed in message bodies	Remove	☐
Email addresses	Headers, quoted replies, CC lists, CRM contacts	Consistent placeholder	☐
Social Security, tax and ID numbers	HR files, invoices, scanned forms	Remove	☐
Medical record and health plan numbers	Clinical systems, billing email, claim forms	Remove, or a code not built from the number	☐
Account, member and card numbers	Billing, CRM, payment notes, support tickets	Remove; keep card data out	☐
Certificate and license numbers	Professional licences, permits, driver's licences	Remove	☐
Vehicle IDs and plates	Dispatch notes, fleet records, claims	Remove	☐
Device serial numbers	IT tickets, scan headers, implant and pacemaker notes	Remove	☐
Web addresses and IP addresses	Personal sites, login logs, email headers	Remove	☐
Biometrics, photos, voices	Profile photos, ID scans, call audio, face photos	Keep out; use transcripts	☐
Any other unique number or code	Customer IDs, employee numbers, case numbers	Replace with a new code	☐

With chapter 4's dates and places, these are HIPAA's 18 Safe Harbor identifiers, 45 CFR 164.514(b)(2).

Placeholders and the key

Before

Hi Maria Lopez, Dave will call you on 555-0142 about account 88210.

After

Hi `customer 118`, `employee 632` will call you on `[phone]` about account `account 12`.

- [ ] **Give each person one placeholder everywhere,** in every system, including nicknames, initials and misspellings of the same name.
- [ ] **Build codes from nothing about the person.** A hash of an email or a number can often be matched back, and HIPAA allows a re-identification code only if it is not derived from the person's information (164.514(c)).
- [ ] **Store the key apart from the data,** encrypted, with a named list of who can open it, or destroy it.
- [ ] **Destroy the key if the data must be anonymous under GDPR,** and write down the date.

4 Dates, places and ages

Safe Harbor keeps only the year and the first three digits of a ZIP code. Anything finer needs an expert.

Dates

Option	What it does	Safe Harbor?	Done
Year only	Birth, visit, admission, discharge, death and service dates cut to the year.	Yes	☐
Shift per person	Every date for one person moves by the same random number of days, so the gaps between events survive.	No, Expert Determination	☐
Days from a start date	Dates become "day 0, day 14, day 90" from a first visit or a contract start.	No, Expert Determination	☐
Month and year	Day removed, month kept.	No, Expert Determination	☐

- [ ] **Check the dates outside the date fields:** message bodies, email headers, file names, scan headers and timestamps carried over from the source system.
- [ ] **Ages over 89 become "90 or older",** and any date or year that would reveal that age goes too.
- [ ] **Write down the method,** and keep the shift values with the key, or destroy them.

ZIP codes and places

What you have	Safe Harbor action	Done
Street, city, county, 5-digit ZIP	Remove.	☐
First three ZIP digits, area over 20,000 people	Keep.	☐
First three ZIP digits, area of 20,000 or fewer	Change to 000. Under the 2000 Census these were 036, 059, 063, 102, 203, 556, 692, 790, 821, 823, 830, 831, 878, 879, 884, 890 and 893. Check current Census data.	☐
State	Keep.	☐

Places also hide in clinic letterheads, office names, ward and branch names, job site addresses and area codes.

Worksheet: identifier log

One line per kind of identifier. This becomes the core of your record in chapter 8.

Identifier	Found in	Action	How many	Checked by, date
Example: phone numbers	Signatures, ticket notes	Removed	41,200	Second reviewer, 3 March

Most personal details in company data sit in places no field labels, so people read the text after software has flagged what it can.

Messages and tickets

- [ ] **Email signatures** with name, title, direct line and office address
- [ ] **Quoted replies and forwards,** which repeat the original sender's details further down
- [ ] **Chat @mentions, display names and user IDs**
- [ ] **Ticket notes and internal comments,** where staff paste phone numbers, addresses and order details
- [ ] **Clinical and case notes:** referring doctors, co-signers, relatives, and dictated names

Files

- [ ] **File names,** such as "Lopez_Maria_claim_2023.pdf"
- [ ] **Document properties:** author, last editor, company, tracked changes and comments
- [ ] **Scanned PDFs and images:** the text has to be read from the image before it can be checked
- [ ] **Photo metadata:** GPS location, device, date taken
- [ ] **Medical scan (DICOM) headers:** patient name, ID, birth date, institution, device serial and scan time. Exporting the image to JPEG or PNG does not clean the original header, and copies of it can survive in logs and export folders.
- [ ] **Call audio:** a voice can identify someone, so keep audio out and send cleaned transcripts

Your company and your clients

Data can point back to your company, and through it to the people you serve. NIST SP 800-188 counts that risk too.

- [ ] **Replace your company name, domains and product names** if the recipient should not know the source
- [ ] **Replace client and supplier names, project code names, invoice and job numbers**
- [ ] **Check for unique events:** a lawsuit, a fire, a merger or a news story that names the company in one line

Fields that are harmless alone can single someone out together. Count how many records share each combination.

These are quasi-identifiers: age, sex, ZIP area, job title, a rare diagnosis, an unusual event. A data set where every combination appears at least k times meets k-anonymity. Many teams require k of 5 or 10; under Expert Determination the expert sets the bar.

- [ ] **List the quasi-identifiers you kept.**
- [ ] **Count the records for every combination** of them. The smallest count is your k.
- [ ] **Flag combinations under your threshold,** and every record that is unique (k of 1).
- [ ] **Fix them** with one of the methods below, then count again.
- [ ] **Check the sensitive field inside each group.** If all five people in a group share one diagnosis, the group reveals it (l-diversity).
- [ ] **Write down the threshold, the final k, and why.**

Example count

Age band	Sex	ZIP area	Job title	Records (k)	Risk	Fix
30 to 39	F	021	Accountant	41	Low	None
50 to 59	M	606	Site foreman	9	Low	None
60 to 69	F	840	Chief financial officer	2	High	Title to "executive"
90 or older	M	000	Retired	1	High	Suppress the record

Fixes

Method	What it does	Use it for
Generalise	Exact value to a range or broader group	Ages to bands, titles to job families, diagnosis codes to their first three characters
Suppress	Remove the value, or the whole record	Unique records, conditions with fewer than 5 cases
Top or bottom code	Cap extreme values	Very high salaries or claim amounts, very large companies
Swap	Exchange a value between similar records	Under Expert Determination, when totals must stay right

7 Test it

Run the automatic checks, then have someone who did not clean the data try to identify people in it.

Automatic checks

Check	Passes when	Done
Identifier scan	No names, emails, phone numbers, ID numbers, IP addresses or web addresses are found in any field or text	☐
Dates	No full dates remain where only years are allowed, including in text and file names	☐
ZIP codes	Only three digits remain, and restricted areas read 000	☐
Ages	No age over 89; those read "90 or older"	☐
Placeholders	One person has one placeholder across every system	☐
File metadata	Author, editor, comments, tracked changes and photo GPS are empty	☐
Combinations	The smallest group meets the k you set in chapter 6	☐

People's review

- ☐ **Read a random sample of records,** at least 100 or 10%, whichever is larger. Read every record when the stakes are high, such as health data or a regulator's review.
- ☐ **Run the motivated intruder test** (UK ICO guidance): could a determined person with the internet, public records and social media identify anyone or your company?
- ☐ **Have a second person review** anything the first one was unsure about.
- ☐ **Send failures back to chapter 3 to 6,** fix them across the whole data set, and test again.

Does the data still do its job?

- ☐ **Conversations still read:** you can tell who said what, and that two tickets came from the same customer
- ☐ **Order of events survives** where the recipient needs it
- ☐ **Write down what was lost** to privacy, so the recipient knows

8 Record, share and review

Keep a written record of every step, put the limits in the contract, and check again when anything changes.

The record

- [ ] **Laws applied,** and why
- [ ] **Every kind of identifier found,** and whether it was removed, coded or cut back
- [ ] **Methods and settings:** date method, ZIP list used, k threshold, suppression rules
- [ ] **Test results** and the risk left, in plain words
- [ ] **Who did each step, who reviewed it,** and when
- [ ] **When the raw copies and the key were destroyed**
- [ ] **The expert's signed report,** if you used Expert Determination

Sharing

- [ ] **The contract bans any attempt to re-identify** people or your company
- [ ] **It limits use** to the purpose you wrote down in chapter 1
- [ ] **Anyone the recipient passes data to** signs the same terms (CCPA requires this)
- [ ] **Keep a list of every recipient,** what they received and when

Check again when

- [ ] The data changes, or new records are added
- [ ] A new recipient or a new use comes along
- [ ] New public data appears that could be matched against yours
- [ ] A year has passed since the last check

If someone may have been identified

- [ ] **Stop access** to the data set
- [ ] **Find out what happened** and how far it went
- [ ] **Tell your privacy officer and counsel** the same day; they decide on notices under the law that applies
- [ ] **Fix the method** before anything is shared again

✓ Sign-off

Role	Name	Signature	Date
Data owner			
Privacy or compliance officer			
Person who cleaned the data			
Person who tested it			
Qualified expert (Expert Determination only)			

Want us to do it for you?

We map, clean and test your data to the rules that apply, and hand you the record. Your data travels and sits encrypted on our US server, raw copies are destroyed on a date we agree in writing, and we sign a BAA before any health records reach us. Priced per project, with a time estimate in the quote.

AITrainingDataBroker.com/de-identification



Jimmy Lin, AI Training Data Broker

Fortune 500 data analyst, data engineer and software engineer, trained in data science. Call or text **+1 656 800 9088**.

Sources

- HIPAA Privacy Rule, 45 CFR 164.514(b) and (c); HHS, Guidance Regarding Methods for De-identification of Protected Health Information
- California Civil Code 1798.140(m), deidentified data
- GDPR Recital 26 and Article 4(5); UK ICO, anonymisation guidance and the motivated intruder test
- NIST Special Publication 800-188, De-Identifying Government Datasets
- European Medicines Agency, External guidance on the implementation of Policy 0070 (clinical data publication)
- Latanya Sweeney, Simple Demographics Often Identify People Uniquely, Carnegie Mellon University, 2000

General information, not legal advice. Your counsel confirms which laws apply and what they require. We do not certify data or sign HIPAA Expert Determinations; a qualified expert does.